

Manu Yadav

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Summary

Passionate and results-driven AI/ML professional with hands-on experience in Python, PyTorch, SQL, and cutting-edge GenAI technologies. Proven track record of delivering impactful machine learning and deep learning solutions through both company-led and personal projects. Eager to innovate at the intersection of data, intelligence, and real-world problem solving.

Education

Chandigarh University

Jul 2018 – Jul 2022

BE in Computer Science

- **CGPA:** 7.32/10.0
- **Coursework:** Applied Mathematics, Algorithms, Data Structures, Databases, Machine Learning, Programming Languages

Experience

Globallogic

Gurugram, Haryana

Trainee Software Engineer → Associate Software Engineer → Software Engineer → Senior Software Engineer

Jul 2022 – Present

Senior Software Engineer

Jan 2024 – Present

- **Generative AI Projects:** Led the development of "*Content Factory*", a generative AI based system to produce optimized ad content. The solution is now used by over **100 Google sales teams** to boost client *ad performance* and *conversion rates*.
- **Automation at Scale:** Automated the **Brand Lift Study (BLS)** workflow using **Apps Script** and **Google Cloud Platform (GCP)**, resulting in estimated annual savings of **\$1 million** by eliminating manual processes.
- **Forecasting & Competitive Analysis:** Executed multiple *data-driven projects* involving **forecasting models** and **competitor comparisons** to support strategic decision-making and *sales enablement*.

Software Engineer

Jul 2023 – Dec 2023

- **Feedback Clustering Solution:** Built a machine learning pipeline to cluster tens of thousands of user feedback entries into *20–50 concise summaries*, significantly reducing manual workload and enhancing analysis efficiency for *Googlers*.
- **Interactive Reporting System:** Developed a **Retrieval-Augmented Generation (RAG)** system for *interactive data reporting*, enabling users to query and explore analytics reports in a *conversational, dynamic* manner.
- **Machine Learning and Data Science:** Applied **ML models** to extract *actionable insights*, predict ad performance, and recommend *optimal bidding and investment strategies* for advertisers. Conducted extensive **data analysis** to optimize ad spend and campaign timing.

Associate Software Engineer

Dec 2022 – Jul 2023

- **SQL Data Extraction:** Queried and extracted structured **Google Ads** data from internal databases to support downstream *analytics workflows*.
- **Performance Reporting:** Built automated, data-rich performance reports revealing key trends and **ROI**, empowering Google's sales teams with **actionable insights**.
- Developed domain expertise in **Google Ads (AdWords)**, enabling sharper *ad strategy formulation* and improved client impact.

Trainee Software Engineer

Jul 2022 – Nov 2022

- Completed intensive hands-on training in **SQL** and **PL/SQL**.
- Secured **1st rank** in the batch project by designing an efficient supply chain database.

Publications

Factors of Unhappiness in India *International Journal of Management Research and Social Science (IJMRSS)*

Dec 2020

ISSN 2394-6415 [🔗](#)

Projects

Image Clustering using CNNs and K-Means

[Github Link](#) [🔗](#)

- Built an unsupervised image clustering system inspired by Google Photos and Apple Photos' intelligent photo grouping.
- **Feature Extraction:** Used pre-trained VGG16 Convolutional Neural Network to extract high-dimensional image features.
- **Dimensionality Reduction:** Applied Principal Component Analysis (PCA) to reduce feature dimensionality while preserving variance.
- **Clustering:** Implemented K-Means clustering, with optimal cluster count determined via the `elbow method`.
- Grouped images into categories such as *cats/dogs, humans, and food*, enabling content-based image retrieval and automatic photo organization.

Fine-Tuning Gemini-Pro LLM on Jawaharlal Nehru's Writings

[Github Link](#) [🔗](#)

- Fine-tuned Google's **Gemini-Pro** Large Language Model on a custom dataset curated from Jawaharlal Nehru's *The Discovery of India*, enabling accurate and context-aware responses emulating the historical figure's voice and tone.
- Employed OpenAI API and Google Cloud Platform (GCP) for dataset preprocessing, model orchestration, and evaluation, showcasing expertise in modern AI and cloud workflows.
- Achieved high accuracy and relevance in simulated historical QA, highlighting potential applications in educational chatbots and heritage preservation.

Transformer (Nano-GPT) from Scratch in PyTorch

[Github Link](#) [🔗](#)

- Built a GPT-style decoder-only transformer from scratch using PyTorch, inspired by Andrej Karpathy's "Makemore" series — no code copied, deepened internal understanding of LLMs.
- Implemented key components: token/position embeddings, LayerNorm, masked multi-head self-attention, MLP, residual connections, and final linear projection head.
- Trained a mini-GPT (5.8M parameters) on the complete works of Shakespeare using character-level tokenization (vocab size: 65), achieving a final loss of 0.8154 (overfitted).
- Trained on Google Colab (NVIDIA T4 GPU) for 15 minutes / 5,000 iterations using context size of 128 and AdamW optimizer with cross-entropy loss.

AI-Powered Bug Resolution Assistant

[Github Link](#) [🔗](#)

- Built an AI-driven assistant using **Vertex AI** and **Pinecone** to retrieve similar historical bugs and provide summarized solutions, significantly reducing bug resolution time.
- Integrated support for custom dataset uploads and used **Gemini AI** for context-aware solution summarization; deployed the tool on **Google App Engine** for scalable access.
- Selected among the top 10 out of 200+ entries in *GlobalLogic's Hackathon 2024*; currently under evaluation for company-wide adoption.

Technologies

Languages: Python, SQL, C++, C, PL/SQL

Frameworks: PyTorch, TensorFlow, LangChain, HuggingFace

Machine Learning & Data Science: Traditional Machine Learning, Deep Learning, Time Series Analysis, Feature Engineering, Model Evaluation, Data Preprocessing, Data Visualization

Generative AI: Transformers, LLM Fine-Tuning, Prompt Engineering, Retrieval-Augmented Generation (RAG), OpenAI API, Vertex AI, Vector Database